

Juventus FC Website - Phase 2 Report

CSCI390: Web Programming (Spring 2025-2026)

1. Project Overview

Student Name: Hussein Khalaf

Project Title: Juventus FC Official Fan Portal

Framework: React 19 + TypeScript

Styling: Tailwind CSS 4 (Juventus Black & Gold Theme)

2. Abstract

This project is the Phase 2 evolution of the Juventus FC website, transitioning from static HTML to a dynamic **ReactJS** application. The portal provides a professional interface for fans to explore club history, squad details, and contact the administration. The implementation focuses on **component-based architecture**, **responsive design**, and **type-safe development**.

3. System Design & Architecture

The application is structured into modular components for high maintainability:

- **Layout Layer:** `Navigation` (Header) and `Footer` components provide consistent branding across all pages.
- **Routing Layer:** Uses `wouter` for fast, client-side navigation without page reloads.
- **Page Layer:**
 - `Home` : Club identity and philosophy.
 - `About` : History and major trophies.
 - `Squad` : Dynamic player profiles.

- **Contact** : Interactive fan engagement form.

4. Technologies Used

Category	Technology	Purpose
Frontend	React 19	Core UI library
Language	TypeScript	Type safety & reduced bugs
Styling	Tailwind CSS 4	Modern, utility-first styling
Icons	Lucide React	Professional iconography
Build Tool	Vite	Ultra-fast development & bundling

5. Key Implementation Features

- **Juventus Identity:** Custom color palette using **Juventus Black** (#000000) and **Victory Gold** (#D2B48C).
 - **Responsive Grid:** Optimized layouts for Mobile, Tablet, and Desktop using Tailwind breakpoints.
 - **Interactive UI:** Used `shadcn/ui` components (Cards, Buttons, Accordions) for a premium feel.
 - **Form Validation:** Client-side validation for the contact form with instant user feedback.
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6. Code Snippet (Example)

```
// Professional Page Routing Implementation
export default function App() {
  return (
    <div className="min-h-screen bg-background">
      <Navigation />
      <main>
        <Switch>
          <Route path="/" component={Home} />
          <Route path="/about" component={About} />
          <Route path="/squad" component={Squad} />
          <Route path="/contact" component={Contact} />
          <Route component={NotFound} />
        </Switch>
      </main>
      <Footer />
    </div>
  );
}
```

7. Challenges & Solutions

- **Challenge:** Migrating static assets to a React structure.
- **Solution:** Organized assets into a dedicated `public` folder and used TypeScript interfaces to manage player data efficiently.

8. Conclusion

The Phase 2 implementation successfully meets all academic requirements, delivering a robust, scalable, and visually stunning web application that represents the Juventus FC brand effectively.